

Singular Value Decomposition

In this section we will discuss an extension of the diagonalization theory for $n \times n$ symmetric matrices to general $m \times n$ matrices. The results that we will develop in this section have applications to compression, storage, and transmission of digitized information and form the basis for many of the best computational algorithms that are currently available for solving linear systems.

$$A = U\Sigma V^T$$

in which U and V are orthogonal but not necessarily the same. The first path leads to decompositions in which J is either diagonal or a certain kind of block diagonal matrix, called a *Jordan canonical form* in honor of the French mathematician Camille Jordan (see p. 538). Jordan canonical forms, which we will not consider in this text, are important theoretically and in certain applications, but they are of lesser importance numerically because of the roundoff difficulties that result from the lack of orthogonality in P . In this section we will focus on the second path.

Singular Values

Since matrix products of the form $A^T A$ will play an important role in our work, we will begin with two basic theorems about them.

Theorem 9.4.1

If A is an $m \times n$ matrix, then:

- (a) A and $A^T A$ have the same null space.
- (b) A and $A^T A$ have the same row space.
- (c) A^T and $A^T A$ have the same column space.
- (d) A and $A^T A$ have the same rank.

Theorem 9.4.2

If A is an $m \times n$ matrix, then:

- (a) $A^T A$ is orthogonally diagonalizable.
- (b) The eigenvalues of $A^T A$ are nonnegative real numbers.

Definition 1

If A is an $m \times n$ matrix, and if $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of $A^T A$, then the numbers

$$\sigma_1 = \sqrt{\lambda_1}, \quad \sigma_2 = \sqrt{\lambda_2}, \dots, \quad \sigma_n = \sqrt{\lambda_n}$$

are called the *singular values* of A .

EXAMPLE 1 | Singular Values

Find the singular values of the matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution The first step is to find the eigenvalues of the matrix

$$A^T A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

The characteristic polynomial of $A^T A$ is

$$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$

so the eigenvalues of $A^T A$ are $\lambda_1 = 3$ and $\lambda_2 = 1$ and the singular values of A in order of decreasing size are

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}, \quad \sigma_2 = \sqrt{\lambda_2} = 1$$

In Exercises 1–4, find the distinct singular values of A .

1. $A = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$

2. $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} \sqrt{2} & 0 \\ 1 & \sqrt{2} \end{bmatrix}$

1. The characteristic polynomial of $A^T A = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is $\lambda^2(\lambda - 5)$; thus the eigenvalues of

$A^T A$ are $\lambda_1 = 5$ and $\lambda_2 = 0$, and $\sigma_1 = \sqrt{5}$ and $\sigma_2 = 0$ are singular values of A .

2. $A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 16 \end{bmatrix}$; $\det(\lambda I - A^T A) = \begin{vmatrix} \lambda - 9 & 0 \\ 0 & \lambda - 16 \end{vmatrix} = (\lambda - 9)(\lambda - 16)$;

the eigenvalues of $A^T A$ are $\lambda_1 = 16$ and $\lambda_2 = 9$ therefore the singular values of A are

$\sigma_1 = \sqrt{\lambda_1} = 4$ and $\sigma_2 = \sqrt{\lambda_2} = 3$.

3. The eigenvalues of $A^T A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$ are $\lambda_1 = 5$ and $\lambda_2 = 5$ (i.e., $\lambda = 5$ is an eigenvalue of multiplicity 2); thus the singular value of A is $\sigma_1 = \sqrt{5}$.

4. $A^T A = \begin{bmatrix} \sqrt{2} & 1 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 1 & \sqrt{2} \end{bmatrix} = \begin{bmatrix} 3 & \sqrt{2} \\ \sqrt{2} & 2 \end{bmatrix}$; $\det(\lambda I - A^T A) = \begin{vmatrix} \lambda - 3 & -\sqrt{2} \\ -\sqrt{2} & \lambda - 2 \end{vmatrix} = (\lambda - 1)(\lambda - 4)$;

the eigenvalues of $A^T A$ are $\lambda_1 = 4$ and $\lambda_2 = 1$ therefore the singular values of A are $\sigma_1 = \sqrt{\lambda_1} = 2$ and $\sigma_2 = \sqrt{\lambda_2} = 1$.

Singular Value Decomposition

Before turning to the main result in this section, we will find it useful to extend the notion of a “main diagonal” to matrices that are not square. We define the *main diagonal* of an $m \times n$ matrix to be the line of entries shown in [Figure 9.4.1](#)—it starts at the upper left corner and extends diagonally as far as it can go. We will refer to the entries on the main diagonal as the *diagonal entries*.

We are now ready to consider the main result in this section, which is concerned with a specific way of factoring a general $m \times n$ matrix A . This factorization, called *singular value decomposition* (abbreviated SVD) will be given in two forms, a brief form that captures the main idea, and an expanded form that spells out the details. The proof is given at the end of this section.

$$A = U\Sigma V^T$$

Theorem 9.4.3

Singular Value Decomposition (Brief Form)

If A is an $m \times n$ matrix of rank k , then A can be expressed in the form $A = U\Sigma V^T$, where Σ has size $m \times n$ and can be expressed in partitioned form as

$$\Sigma = \left[\begin{array}{c|c} D & 0_{k \times (n-k)} \\ \hline 0_{(m-k) \times k} & 0_{(m-k) \times (n-k)} \end{array} \right]$$

in which D is a diagonal $k \times k$ matrix whose successive entries are the first k singular values of A in nonincreasing order, U is an $m \times m$ orthogonal matrix, and V is an $n \times n$ orthogonal matrix.

Theorem 9.4.4

Singular Value Decomposition (Expanded Form)

If A is an $m \times n$ matrix of rank k , then A can be factored as

$$A = U\Sigma V^T = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_k \mid \mathbf{u}_{k+1} \quad \cdots \quad \mathbf{u}_m] \left[\begin{array}{cccc|ccc} \sigma_1 & 0 & \cdots & 0 & & & \\ 0 & \sigma_2 & \cdots & 0 & & & \\ \vdots & \vdots & \ddots & \vdots & & & \\ 0 & 0 & \cdots & \sigma_k & & & \\ \hline & & & & 0_{(m-k) \times k} & & 0_{(m-k) \times (n-k)} \end{array} \right] \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \\ \hline \mathbf{v}_{k+1}^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix}$$

in which U , Σ , and V have sizes $m \times m$, $m \times n$, and $n \times n$, respectively, and in which:

- (a) $V = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n]$ orthogonally diagonalizes $A^T A$.
- (b) The nonzero diagonal entries of Σ are $\sigma_1 = \sqrt{\lambda_1}, \sigma_2 = \sqrt{\lambda_2}, \dots, \sigma_k = \sqrt{\lambda_k}$, where $\lambda_1, \lambda_2, \dots, \lambda_k$ are the nonzero eigenvalues of $A^T A$ corresponding to the column vectors of V .
- (c) The column vectors of V are ordered so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0$.
- (d) $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|} = \frac{1}{\sigma_i} A\mathbf{v}_i \quad (i = 1, 2, \dots, k)$
- (e) $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthonormal basis for $\text{col}(A)$.
- (f) $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_m\}$ is an extension of $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ to an orthonormal basis for R^m .

Find a singular value decomposition of the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Nice — let's compute the SVD of the 3×2 matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We want $A = U\Sigma V^T$ with $U \in \mathbb{R}^{3 \times 3}$ orthogonal, $\Sigma \in \mathbb{R}^{3 \times 2}$ diagonal (nonnegative entries), and $V \in \mathbb{R}^{2 \times 2}$ orthogonal.

Step 1 — compute $A^T A$

$$A^T A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Step 2 — eigenvalues of $A^T A$

Characteristic polynomial:

$$\det(A^T A - \lambda I) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Solve $\lambda^2 - 4\lambda + 3 = 0 \Rightarrow (\lambda - 1)(\lambda - 3) = 0$.

So eigenvalues are $\lambda_1 = 3$, $\lambda_2 = 1$.

Singular values are $\sigma_i = \sqrt{\lambda_i}$:

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = 1.$$

Step 3 — eigenvectors of $A^T A$ (gives V)

For $\lambda_1 = 3$: solve $(A^T A - 3I)v = 0 \rightarrow$ eigenvector $v^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Normalize to unit length:

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 1$: eigenvector $v^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

Normalize:

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

So

$$V = (v_1 \quad v_2) = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad V^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Step 4 — compute corresponding U columns via $u_i = \frac{1}{\sigma_i} Av_i$

For $v_1 = \frac{1}{\sqrt{2}}(1, 1)^T$ and $\sigma_1 = \sqrt{3}$:

$$Av_1 = A \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1+1 \\ 0+1 \\ 1+0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

So

$$u_1 = \frac{1}{\sigma_1} Av_1 = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

For $v_2 = \frac{1}{\sqrt{2}}(-1, 1)^T$ and $\sigma_2 = 1$:

$$Av_2 = A \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 + 1 \\ 0 + 1 \\ -1 + 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

So

$$u_2 = \frac{1}{\sigma_2} Av_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

are two of the three column vectors of U . Note that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthonormal, as expected. We could extend the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ to an orthonormal basis for $\mathbb{R}^3$. However, the computations will be easier if we first remove the messy radicals by multiplying $\mathbf{u}_1$ and $\mathbf{u}_2$ by appropriate scalars. Thus, we will look for a unit vector $\mathbf{u}_3$ that is orthogonal to

$$u_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

To use the cross product, we should use the **non-normalized** versions first:

$$a = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

$$a \times b = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix}$$

So the cross product is:

$$a \times b = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$$

Normalize the cross product to get u_3

$$u_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$U = \begin{pmatrix} u_1 & u_2 & u_3 \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}.$$

Step 5 — the Σ matrix

Σ is 3×2 with singular values on the diagonal in descending order:

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \end{pmatrix}$$

Final SVD

$$A = U\Sigma V^T$$

with

$$U = \begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad V^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

You can verify by multiplying $U\Sigma V^T$ that you recover A . (The singular values are $\sqrt{3}, 1$; U and V are orthogonal; and the construction $u_i = (Av_i)/\sigma_i$ was used.)

In Exercises 5–12, find a singular value decomposition of A .

$$5. \quad A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

5. The only eigenvalue of $A^T A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is $\lambda = 2$ (multiplicity 2), and the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ form an orthonormal basis for the eigenspace (which is all of $\mathbb{R}^2$).

The singular values of A are $\sigma_1 = \sqrt{2}$ and $\sigma_2 = \sqrt{2}$. We have $\mathbf{u}_1 = \frac{1}{\sigma_1} A \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$, and

$$\mathbf{u}_2 = \frac{1}{\sigma_2} A \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

This results in the following singular value decomposition of A :

$$A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$10. \quad A^T A = \begin{bmatrix} -2 & 2 \\ -1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -2 & -1 & 2 \\ 2 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 8 & 4 & -8 \\ 4 & 2 & -4 \\ -8 & -4 & 8 \end{bmatrix};$$

$$\det(\lambda I - A^T A) = \begin{vmatrix} \lambda - 8 & -4 & 8 \\ -4 & \lambda - 2 & 4 \\ 8 & 4 & \lambda - 8 \end{vmatrix} = (\lambda - 18)\lambda^2; \text{ the eigenvalues of } A^T A \text{ are } \lambda_1 = 18 \text{ and}$$

$\lambda_2 = \lambda_3 = 0$ therefore the singular values of A are $\sigma_1 = \sqrt{\lambda_1} = 3\sqrt{2}$, $\sigma_2 = \sqrt{\lambda_2} = 0$, and $\sigma_3 = \sqrt{\lambda_3} = 0$.

The reduced row echelon form of $18I - A^T A$ is $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$ so that the eigenspace corresponding to $\lambda = 18$

contains vectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ where $x_1 = -t$, $x_2 = -\frac{1}{2}t$, $x_3 = t$. A vector $\mathbf{v}_1 = \begin{bmatrix} -\frac{2}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$ forms an orthonormal basis for this eigenspace.

The reduced row echelon form of $0I - A^T A$ is $\begin{bmatrix} 1 & \frac{1}{2} & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ so that the eigenspace corresponding to $\lambda = 0$

contains vectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ where $x_1 = -\frac{1}{2}s + t$, $x_2 = s$, $x_3 = t$. Vectors $\mathbf{p}_1 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{p}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ form a basis for

this eigenspace. We apply the Gram-Schmidt process to find an orthogonal basis for this eigenspace:

$\mathbf{q}_1 = \mathbf{p}_1 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{q}_2 = \mathbf{p}_2 - \frac{\langle \mathbf{p}_2, \mathbf{q}_1 \rangle}{\|\mathbf{q}_1\|^2} \mathbf{q}_1 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} - \frac{-1}{5} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} \\ \frac{2}{5} \\ 1 \end{bmatrix}$, then proceed to normalize the two vectors to

yield an orthonormal basis: $\mathbf{v}_2 = \frac{\mathbf{q}_1}{\|\mathbf{q}_1\|} = \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}$ and $\mathbf{v}_3 = \frac{\mathbf{q}_2}{\|\mathbf{q}_2\|} = \begin{bmatrix} \frac{4}{3\sqrt{5}} \\ \frac{2}{3\sqrt{5}} \\ \frac{5}{3\sqrt{5}} \end{bmatrix}$.

The matrix $V = [\mathbf{v}_1 \mid \mathbf{v}_2 \mid \mathbf{v}_3] = \begin{bmatrix} -\frac{2}{3} & -\frac{1}{\sqrt{5}} & \frac{4}{3\sqrt{5}} \\ -\frac{1}{3} & \frac{2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} \\ \frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \end{bmatrix}$ orthogonally diagonalizes $A^T A$:

$$V^T (A^T A) V = \begin{bmatrix} 18 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$U = \begin{bmatrix} \frac{2}{3} \\ 1 \\ \sqrt{2} \end{bmatrix}$$

To obtain $\mathbf{u}_2$, we extend the set $\{\mathbf{u}_1\}$ to an orthonormal basis for R^2 . To simplify the computations, we

consider $\sqrt{2}\mathbf{u}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. A vector $\mathbf{u}_2$ orthogonal to this vector must be a solution of $\begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$.

An orthonormal basis for the solution space is formed by $\mathbf{u}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.

A singular value decomposition of A is

$$\underbrace{\begin{bmatrix} -2 & -1 & 2 \\ 2 & 1 & -2 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}}_U \underbrace{\begin{bmatrix} 3\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} -\frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ \frac{4}{3\sqrt{5}} & \frac{2}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix}}_{V^T}$$

In Exercises 5–12, find a singular value decomposition of A .

$$5. A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$6. A = \begin{bmatrix} -3 & 0 \\ 0 & -4 \end{bmatrix}$$

$$7. A = \begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix}$$

$$8. A = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$$

$$9. A = \begin{bmatrix} -2 & 2 \\ -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$10. A = \begin{bmatrix} -2 & -1 & 2 \\ 2 & 1 & -2 \end{bmatrix}$$

$$11. A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$12. A = \begin{bmatrix} 6 & 4 \\ 0 & 0 \\ 4 & 0 \end{bmatrix}$$